Adaptive Financial Advisor using Reinforcement Learning (DQN)

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***Abstract*—This paper presents a framework based on reinforcement learning for autonomous financial decision-making using Deep Q-Learning (DQN). A custom financial environment was developed to simulate monthly income, expenses, investments, and inflation. The agent learns optimal strategies to allocate wealth among *spending*, *saving*, and *investing* actions based on a multi-objective reward function balancing growth, user satisfaction, and financial stress. The results show that the trained agent consistently increases total assets in multiple episodes, demonstrating the potential of reinforcement learning for adaptive and personalized financial planning.**

***Index Terms*—Reinforcement Learning, Deep Q-Network, Financial Decision Optimization, AI Finance, Personalized Advisor.**

I. Introduction

Traditional financial planning tools rely on static heuristics or user-defined rules that do not adapt to dynamic market conditions or individual behavioral patterns. With the rise of data-driven decision systems, reinforcement learning (RL) offers a promising paradigm in which an agent can autonomously learn strategies by interacting with an environment and receiving feedback in the form of rewards.

In this work, we develop an RL agent that learns how to optimally manage finances – deciding when to spend, save, or invest – under stochastic market and income conditions. The approach leverages the Deep Q-Learning (DQN) algorithm to approximate long-term expected rewards and adapt its decision policy accordingly.

II. System Architecture

The overall architecture is shown in Fig. 1. The system consists of three main modules: the environment simulator, the RL agent (DQN), and the evaluation engine.

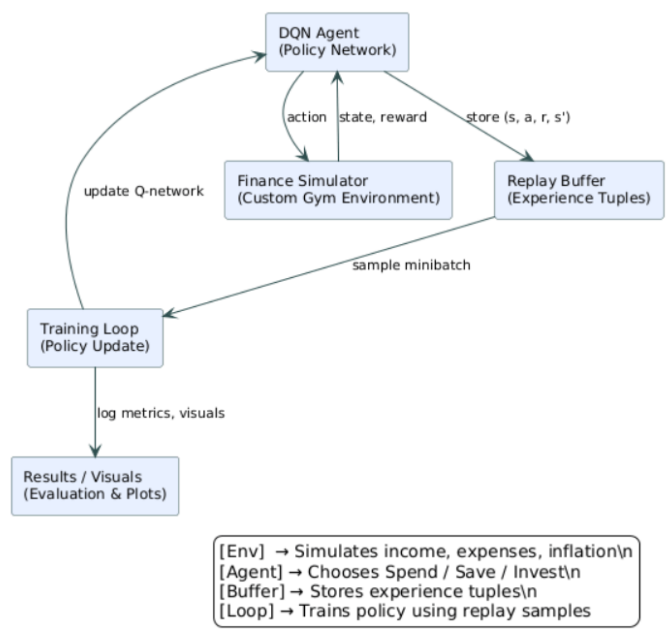

Fig. 1: High-level system architecture of the RL-based financial decision system.

III. Methodology

A. Custom Financial Environment

A Gym-compatible environment was created where each episode represents 36 months of financial activity. At each step, the agent observes a state vector:

$$S_t = [\text{balance}, \text{income}, \text{expenses}, \text{market\_return}, \text{inflation}, \text{risk\_tolerance}, \text{g}$$

The available actions are:

$$A_t \in \{0 : \text{Spend}, 1 : \text{Save}, 2 : \text{Invest}\}$$

The environment introduces stochasticity via normally distributed income, market returns, and inflation.

B. Reward Function

The core of this system is the custom reward function designed to model realistic human-financial trade-offs:

$$R_t = \alpha \cdot G_t + \beta \cdot S_t - \gamma \cdot F_t$$

where:

- G_t: Portfolio growth rate (wealth accumulation)
- S_t: User satisfaction (spending enjoyment and goal progress)
- F_t: Financial stress (cash deficiency and risk penalty)

Hyperparameters (α, β, γ) determine the agent's behavioral bias:

$$\alpha = 0.6, \quad \beta = 0.3, \quad \gamma = 0.1$$

These can be personalized per user to reflect individual risk tolerance and lifestyle preferences.

C. Learning Algorithm

A Deep Q-Network (DQN) with an MLP policy is trained to approximate the Q-function:

$$Q(s, a; \theta) = E[R_t + \gamma \max_{a'} Q(s', a'; \theta')]$$

Experience replay and target networks stabilize training.

The agent was trained for 400k+ timesteps with the following key parameters:

- Learning rate = 1×10^{-4}
- Batch size = 64
- Exploration fraction = 0.4
- Discount factor (γ) = 0.99

IV. Results

The model was evaluated in multiple simulated environments to analyze its learning stability and policy robustness.

A. Training Reward Progression

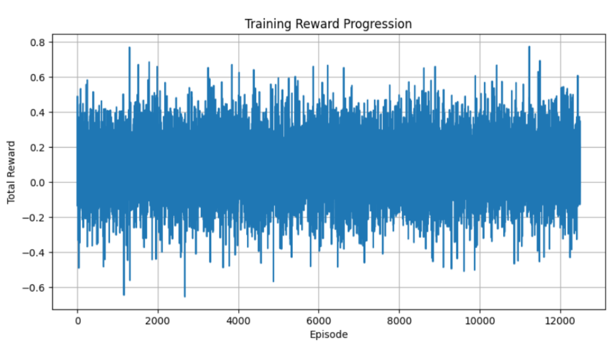

Fig. 2: Training reward progression across episodes.

The training reward curve fluctuates within a stable band after early episodes, indicating convergence. The reward variance is expected due to stochastic market and income conditions.

B. Agent Action Distribution

Fig. 3: Distribution of actions chosen by the trained agent.

The agent demonstrates a conservative policy: prioritizing spending and saving, with limited investment. This reflects a risk-averse balance between satisfaction and long-term growth.

C. Multi-Episode Asset Growth

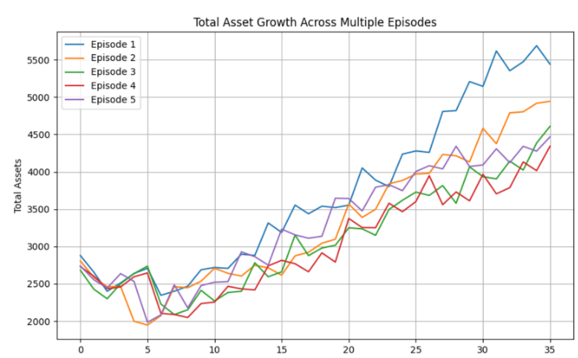

Fig. 4: Total asset growth over multiple episodes.

Despite randomness in market conditions, all episodes exhibit upward trends in total assets, confirming policy generalization and stability.

D. Average Portfolio Growth

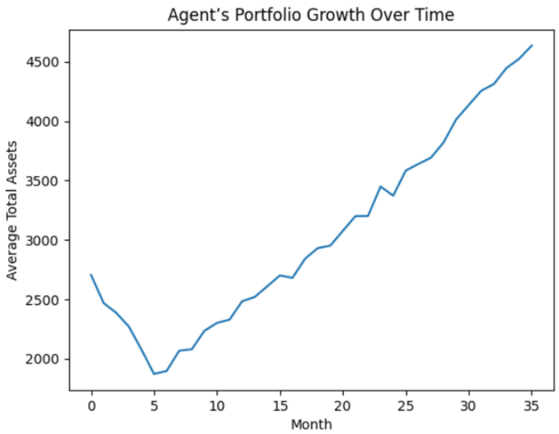

Fig. 5: Average portfolio value over 36 months.

The average trajectory shows consistent growth, indicating that the DQN policy effectively accumulates wealth over time.

V. Discussion

A. Interpretation

The model learns a stable financial strategy that balances spending satisfaction and long-term portfolio growth. Although the agent is risk-averse under the given reward weights, modifying α, β, and γ enables the emulation of different financial personalities (e.g., aggressive investor vs. conservative saver).

B. Real-World Adaptation

In a real-world implementation:

- α, β, γ can be dynamically estimated from user transaction histories, demographics, and risk tolerance questionnaires.
- Real financial data (bank APIs, market APIs) replace simulated variables.
- The trained DQN can be deployed as an API service that provides spend-save-invest recommendations in JSON format.

Example API response:


```
{ "action": "Invest", "confidence": "0.86" }
```


VI. Conclusion

This study demonstrates that reinforcement learning can model and optimize financial decision-making under uncertainty. A DQN-based agent trained in a custom financial environment learned a balanced, conservative policy that yields consistent wealth accumulation across variable market conditions. Future work includes integrating real financial datasets, personalizing reward weights using behavioral finance insights, and deploying the model as an adaptive API for user-facing financial apps.

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